

llmXive Automated Review of Humanoid-GPT: Scaling Data and Structure for Zero-Shot Motion Tracking

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided evidence (tables, text, and citations).

1. Discrepancy in Scaling Claims (Introduction): In the “Science of Scale” paragraph (Introduction), the authors claim their 2B-frame corpus is “over 200x larger than prior tracker training sets.” While this holds true when comparing against older datasets like AMASS (~7.2M frames), the paper explicitly cites and includes **SONIC** (2025) in Table 1 and the text as a related work that scales to **100M frames**.

- Calculation: $2,000\text{M} / 100\text{M} = 20\text{x}$.
- The claim of “200x” is factually incorrect relative to the most recent and relevant baseline (SONIC) presented in the same section. This exaggeration undermines the precision of the “Science of Scale” argument. The text should clarify that the 200x figure applies to *traditional* trackers (e.g., HumanPlus, OmniH2O) rather than the state-of-the-art scaling work (SONIC).

2. Unsupported “First Systematic Evidence” Claim (Introduction): The Introduction states: “Crucially, we provide the first systematic evidence that *video-estimated motion* can materially improve tracking when the model and the training set are scaled appropriately.”

- **Issue:** The manuscript does not present a specific ablation study, figure, or table that isolates “video-estimated motion” as a variable to demonstrate this improvement. The data curation section mentions aggregating datasets (AMASS, MotionMillion, etc.) but does not explicitly detail a controlled experiment comparing “video-estimated” vs. “non-video” data within the 2B corpus to prove the causal link. Without a specific reference to a figure (e.g., “Fig. X”) or a table row showing this specific comparison, the claim of “first systematic evidence” is unsupported by the provided text.

3. Missing Data in Simulation Results (Section 4): In the “Results” paragraph of Section 4, the authors write: “even the best baseline (TCN-L at 56.15mm) lags behind Humanoid-GPT-S (43.25mm) by 30%.”

- **Issue:** Table 2 (`tab:sim_scaling_backbone`) lists a “TCN (8-layer)” with an MPKPE of **79.75mm**, not 56.15mm. Furthermore, there is no row labeled “TCN-L” in the table. The values 56.15mm and 43.25mm do not appear anywhere in the provided tables.
- **Impact:** This suggests either a typo in the text, a missing table row, or a reference to data not included in the current manuscript version. The claim cannot be verified against the provided evidence, rendering the specific quantitative comparison invalid in its current form.

4. Citation Consistency: The paper cites “SONIC” (2025) and “BeyondMimic” (2025) as existing works. While these are future-dated citations (likely arXiv preprints), the internal logic regarding their performance (e.g., SONIC’s 100M frames) is consistent with the text, except for the scaling factor error noted in point #1.

Conclusion: The paper makes strong claims about scaling and novelty that are currently undermined by numerical inconsistencies and missing supporting data in the text. The “200x”

claim is mathematically inconsistent with the cited SONIC baseline, and the specific performance metrics for the “TCN-L” baseline are absent from the tables. These are writing-level fixes (correcting numbers and adding missing references) but are critical for the accuracy of the claims.

Action items.

- **[writing]** In the Introduction (Sec 1, ‘Science of Scale’), the claim that the training set is ‘over 200x larger than prior tracker training sets’ is not supported by the provided data. Table 1 lists SONIC (2025) with 100M frames. $2B / 100M = 20x$, not 200x. The 200x figure likely compares against older baselines (~7-10M frames) but is misleading when SONIC is explicitly cited as a related work in the same section and table. Clarify the baseline used for this comparison.
- **[writing]** In the Introduction (Sec 1, ‘Science of Scale’), the paper claims to provide ‘the first systematic evidence that video-estimated motion can materially improve tracking.’ However, the text does not cite a specific ablation study, figure, or table within the manuscript that isolates ‘video-estimated motion’ as a variable to prove this causal link. The claim appears to be an assertion without the immediate evidentiary support required for a ‘first systematic evidence’ statement.
- **[writing]** In Section 4 (‘Evaluation in Simulation’), the text states: ‘even the best baseline (TCN-L at 56.15mm) lags behind Humanoid-GPT-S (43.25mm) by 30%.’ However, Table 2 (tab:sim_scaling_backbone) does not contain a row for ‘TCN-L’ nor does it list MPKPE values of 56.15mm or 43.25mm. The table lists TCN (8-layer) with MPKPE 79.75mm. The specific numbers and the ‘TCN-L’ variant mentioned in the text are missing from the provided tables, making the claim unverifiable from the current source.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear visual overview of the pipeline stages, but the caption is marred by grammatical errors where the subject is missing, and it references specific reward types not explicitly labeled in the diagram.

Figure 2

The figure effectively visualizes dataset diversity with clear axes and annotations, but the top x-axis label ‘Geometric-std’ conflicts with the caption’s ‘gstd’, and the label for the largest bubble is ambiguously positioned.

Figure 3

The figure effectively demonstrates the robot’s zero-shot capabilities through clear visual examples of various tasks. However, the caption contains grammatical errors with missing nouns, and the task labels on the image suffer from poor contrast.

Figure 4

The bar chart effectively visualizes latency differences but the caption’s claim of ‘5 times faster’ is imprecise relative to the specific data points shown, and the separation of the TWIST baseline by a dashed line without explanation creates ambiguity in the comparison.

Figure 5

Figure 5 effectively demonstrates the system’s zero-shot generalization capabilities through a grid of real-world video frames showing the humanoid tracking diverse human motions. The visual evidence aligns perfectly with the caption’s claim of handling complex and high-dynamic tasks, including specific dance styles.

Figure 6

Figure 6 effectively visualizes the trend of performance improvement with data scaling, but the y-axis lacks a numerical scale and tick marks, relying solely on data labels which limits the ability to assess the magnitude of change.

Figure 7

The figure displays training loss curves over training steps, which contradicts the caption’s claim of a ‘Model Scalability Comparison’ that should typically relate performance to model size or parameters. Additionally, the y-axis label formatting is poor.

Figure 8

The figure presents a pie chart of data sources but suffers from a mathematical inconsistency where the percentages sum to 99% without an ‘Other’ category. Additionally, the caption is insufficiently descriptive, failing to define the categories or the total dataset size.

Figure 9

Figure 9 presents parameter sensitivity analyses rather than clear ablation studies, lacks y-axis scaling, and omits error bars, reducing its scientific rigor and interpretability.

Figure 10

Figure 10 is a scatter plot with a legend but no caption or axis labels, rendering the visualization scientifically unintelligible and impossible to verify against the paper’s claims.

Action items.

- **[writing]** Figure 1: The caption contains multiple grammatical errors and missing nouns, reading ‘Overview of .’ and ‘The resulting can take...’, which obscures the subject (Humanoid-GPT).
- **[writing]** Figure 1: The caption describes ‘keypoint-level rewards’ for stage (b), but the diagram only labels the loop as ‘RL’ without specifying the reward type.
- **[science]** Figure 2: The top x-axis is labeled ‘Geometric-std’ but the caption defines the horizontal axis as ‘gstd’; the label should be updated to match the caption’s terminology.
- **[writing]** Figure 2: The text ‘Our Training data’ is placed below the large brown bubble rather than inside it or connected by a leader line, which creates ambiguity about whether the label refers to the bubble or the empty space below.
- **[writing]** Figure 3: The caption contains a grammatical error and missing noun (‘Overview of .’, ‘Real-world experiments for our .’), likely due to a missing model name placeholder.
- **[writing]** Figure 3: The text labels for specific tasks (e.g., ‘Sit on sofa’, ‘Play with cat’) are rendered in a low-contrast light blue font that is difficult to read against the background.
- **[science]** Figure 4: The claim ‘5 times faster’ is unsupported by the data shown; the fastest method (C++ COMM, 0.39 ms) is only ~8.5x faster than TWIST (3.32 ms), while the labeled ‘final optimization’ (TensorRT & Cache, 0.58 ms) is only ~5.7x faster. The comparison baseline (TWIST) is separated by a dashed line and not grouped with the optimization methods, making the direct comparison ambiguous.
- **[writing]** Figure 4: The y-axis labels are cluttered and inconsistent; ‘C++ COMM’ and ‘TWIST CPU ONNX’ are split across two lines while others are not, and the dashed line separating TWIST is not explained in the caption.
- **[science]** Figure 6: The y-axis label ‘Zero-Shot MPJPE’ is present, but the caption ‘Data Scaling up Curve on Zero-shot Performance’ is vague; it should explicitly state that lower MPJPE indicates better performance to ensure the downward trend is interpreted correctly.
- **[writing]** Figure 6: The y-axis has a break symbol (zigzag) but no numerical scale or tick marks are provided, making it impossible to verify the magnitude of the improvement or the spacing between data points.

- **[science]** Figure 7: The plot displays ‘Training Loss’ on the y-axis, but the caption describes it as a ‘Model Scalability Comparison’. Scalability comparisons typically show performance metrics (e.g., accuracy, success rate) or efficiency metrics (e.g., FLOPs, latency) against model size or parameter count, not training loss curves.
- **[science]** Figure 7: The x-axis is labeled ‘Training Steps’ (50K to 200K), which measures training duration rather than model scale (e.g., number of parameters). This contradicts the caption’s claim of a ‘Model Scalability Comparison’.
- **[writing]** Figure 7: The y-axis label ‘Training Loss’ is split across two lines (‘Training’ and ‘Loss’) with excessive vertical spacing, making it difficult to read as a single unit.
- **[science]** Figure 8: The pie chart displays percentages (68%, 18%, 9%, 4%) that sum to 99%, implying a missing 1% slice or rounding error, yet no ‘Other’ category is shown to account for the discrepancy.
- **[writing]** Figure 8: The caption ‘Data distribution visualization’ is too vague; it fails to specify what the categories (PHUMA, Inhouse, etc.) represent (e.g., source of motion data, dataset split) or the total number of samples.
- **[writing]** Figure 8: The label ‘Motion Million’ is ambiguous and likely a typo or truncated name; it is unclear if this refers to a specific dataset or a category of motion.
- **[science]** Figure 9: The figure displays three bar charts showing ‘Success Rate’ vs. ‘Cluster Num’, ‘History Length’, and ‘Envs Num’, but the caption only states ‘Ablation studies for Humanoid-GPT’ without specifying what is being ablated or what the baseline is. The charts show parameter sweeps rather than ablation of specific model components.
- **[writing]** Figure 9: The y-axis lacks a scale or tick marks, making it impossible to judge the magnitude of differences between bars beyond the labeled values. The x-axis labels are rotated and partially illegible in the rendered image.
- **[science]** Figure 9: No error bars or confidence intervals are shown despite presenting experimental results, which limits the ability to assess statistical significance of the observed differences.
- **[fatal]** Figure 10: The figure has no caption, making it impossible to interpret the axes, the meaning of the ‘Blue’, ‘Brown’, and ‘Light blue’ labels, or the nature of the visualization (e.g., t-SNE, UMAP, or raw data).
- **[science]** Figure 10: The plot lacks axis labels and units, preventing any assessment of the feature space dimensions or scale.
- **[science]** Figure 10: The legend is ambiguous; ‘Light blue: All’ suggests an overlay of the other two categories, but the visualization does not clearly distinguish the specific contributions of ‘AMASS’ and ‘LAFAN’ versus the combined set.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating unnecessary barriers for non-specialist readers.

In the **Abstract** and **Section 1**, the term “G1-retargeted” and the standalone “G1” are used without defining the Unitree-G1 humanoid robot. Similarly, “MoCap” is used repeatedly (e.g., Section 1, Section 3.1) without spelling out “motion capture.”

Section 3.1 introduces “PD controller” and “DoF” (degrees of freedom) without definition. While standard in control theory, a general audience requires these expanded. The text also uses “priv.” as a superscript for state ($s_t^{priv.}$) without explicitly stating “privileged” in the immediate context, relying on the reader to infer the meaning of “privileged observations.”

In **Section 3.2**, “DAgger” is introduced as a framework without defining the acronym (Dataset Aggregation). The loss function “SmoothL1Loss” is named but not described.

Section 2 and **Table 1** use “MoE” (Mixture of Experts) without definition.

The **Appendix** (sec/X_suppl.tex) contains “t-SNE” without expansion and a likely typo “video-esti” which should be “video-estimated.” Finally, the **Conclusion** mentions “VLA-style” without defining Vision-Language-Action models.

These instances of undefined jargon and acronyms violate the principle of accessibility for a broader scientific audience. The authors should ensure every acronym is defined at first use and replace shorthand terms with their full names where appropriate.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating unnecessary barriers for non-specialist readers. In the Abstract and Section 1, the term “G1-retargeted” and the standalone “G1” are used without defining the Unitree-G1 humanoid robot. Similarly, “MoCap” is used repeatedly (e.g., Section 1, Section 3.1) without spelling out “motion capture.” Section 3.1 introduces “PD controller” and “DoF” (degrees of freedom) without de

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is generally sound in its high-level narrative: the premise that scaling data and model capacity improves generalization is supported by the presented

ablation studies and the comparison between MLP/TCN and Transformer architectures. The causal link between the “Science of Scale” (2B frames) and the observed “zero-shot” capabilities is well-motivated by the experimental setup.

However, there are specific logical gaps and inconsistencies in the evidence presentation that prevent a full “accept” verdict:

1. **Unsupported Numerical Claims:** In Section 4.1 (“Results”), the authors make specific quantitative claims about a “TCN-L” model trained on 2B tokens (89.05% SR, 56.15mm MPKPE). These specific data points are absent from Table 1 (`tab:sim_scaling_backbone`), which only lists a generic “TCN (8-layer)” at 2M tokens. Without the corresponding table row or figure data for TCN-L at 2B tokens, the conclusion that “data scaling saturates” for TCNs but not for Transformers is not fully supported by the visible evidence. The reader cannot verify the saturation point or the specific performance gap claimed.
2. **Undefined Terms in Equations:** In Section 3.1, Equation 1 defines a reward term $R_{\text{panel}}(t)$. However, the subsequent text describes this term as $R_{\text{panel}}(t)$ (“ $R_{\text{panel}}(t)$ consists of several penalties...”). This typo creates a logical disconnect between the mathematical definition and the textual explanation, leaving the reader to guess if “panel” is a typo for “penal” or a distinct, undefined term.
3. **Vague Scaling Law Derivation:** Section 5 claims to “derive a scaling law” and “quantify the relationship.” However, the text only qualitatively describes “monotonic scaling trends” and “marginal gains decrease.” It does not present the fitted equation, the exponents, or the statistical fit (e.g., R^2) that would be required to logically support the claim of a derived “law” rather than just an observed trend. The reference to Figure 4 (`data_scaling.pdf`) is made, but the text fails to extract the quantitative parameters from that figure to substantiate the “law” claim.

These issues are primarily related to the completeness of the evidence supporting specific causal claims and the precision of the mathematical definitions.

Action items.

- **[writing]** In Section 3.1 (Eq. 1), the reward function includes a term $R_{\text{extpenal}}(t)$, but the text immediately following refers to it as $R_{\text{extpanel}}(t)$. This typo breaks the logical link between the defined equation and the explanation of penalty terms.
- **[science]** In Section 4.1, the text claims ‘TCN-L achieves 89.05% at 2B tokens’ and cites a 56.15mm MPKPE gap. However, Table 1 only reports TCN results at 2M tokens. The specific data for TCN-L at 2B tokens is missing, making the claim of saturation unsupported by visible evidence.
- **[science]** In Section 5, the paper claims to ‘derive a scaling law’ but only provides qualitative descriptions of trends. Without fitted exponents or a mathematical form, the claim of a derived law is not logically supported by the presented analysis.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the provided evidence, particularly regarding the magnitude of generalization, the existence of a formal scaling law, and the specific contribution of data modalities.

First, the abstract and introduction repeatedly use the term “unprecedented” to describe the zero-shot generalization capabilities (Abstract, Sec 1). However, the empirical evidence for this is limited to a small set of four specific dance clips (Table 3) and a few teleoperation demonstrations (Fig 4, 5). The paper does not present a comprehensive zero-shot benchmark covering a wide variety of unseen motion categories, styles, or dynamic regimes that would be necessary to support such a strong superlative claim against existing baselines like SONIC. The evaluation appears cherry-picked to demonstrate success rather than stress-test the limits of generalization.

Second, Section 5 is titled “Scaling Laws” and claims to “derive a scaling law” for humanoid motion tracking. However, the section lacks the mathematical rigor expected of such a claim. The authors present qualitative plots (Fig 6, 7) and state that “marginal gains decrease slightly,” but they do not provide a fitted power-law equation, exponents, or statistical analysis to validate a specific scaling relationship. Without a formal derivation or quantitative fit, describing this as a “scaling law” is an overreach; it is merely an observation of scaling trends.

Third, the authors claim to provide “the first systematic evidence that video-estimated motion can materially improve tracking” (Sec 1, Summary). While the dataset includes video-derived motions, the paper does not isolate this variable. There is no ablation study comparing training on the full 2B corpus versus a subset excluding video-estimated data to quantify the specific marginal gain attributable to this modality. The improvement is conflated with the overall scale of the dataset.

Finally, the claim of “real-time performance” based on a 1.5ms inference latency on an NVIDIA RTX 4090 (Sec 4.5) is misleading in the context of the deployed system. The Unitree-G1 humanoid typically relies on onboard compute (e.g., Jetson Orin) which is significantly less powerful than a desktop 4090. Without reporting latency measurements on the actual onboard hardware, the claim that the system maintains real-time performance in deployment is unsupported.

Action items.

- **[writing]** The paper makes several claims that extend beyond the provided evidence, particularly regarding the magnitude of generalization, the existence of a formal scaling law, and the specific contribution of data modalities. First, the abstract and introduction repeatedly use the term “unprecedented” to describe the zero-shot generalization capabilities (Abstract, Sec 1). However, the empirical evidence for this is limited to a small set of four

specific dance clips (Table 3) and a few teleoperation de

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily in the dedicated “Ethical Considerations” section (sec/6_limitations.tex). While the authors acknowledge potential risks such as physical collisions and misuse, the current disclosure is insufficient for a work involving large-scale human data collection and physical robot deployment.

First, regarding data privacy and consent, the authors claim in sec/6_limitations.tex that “All human motion recordings are obtained with informed consent.” However, the paper aggregates a “large internally captured dataset” (Introduction, sec/1_intro.tex) alongside public datasets. For the internal data, the manuscript lacks specific details regarding the Institutional Review Board (IRB) approval process, the specific consent forms used, and how data anonymization was technically enforced beyond “stripping personal identifiers.” In the context of motion capture, which can be biometrically identifying, a more rigorous description of the consent protocol and IRB oversight is necessary to ensure compliance with ethical research standards.

Second, the deployment of a zero-shot, high-agility controller on a physical humanoid robot introduces significant safety risks. The “Robot Safety” paragraph acknowledges the risk of collision and falls but relies on a generic “emergency stop mechanism.” Given the system’s capability to perform complex, dynamic tasks like “collaboratively carrying boxes” (sec/4_experiments.tex, Fig. show1) in real-time, the review requires a more concrete safety framework. This should include a quantitative risk assessment, specific fail-safe logic (e.g., force-limiting, collision detection thresholds), and a clear protocol for human-in-the-loop supervision during the reported real-world experiments. The current text is too vague to assure that the system is safe for the described applications.

Finally, while the “Misuse Potential” paragraph correctly identifies risks like surveillance, it does not propose specific technical mitigations (e.g., watermarking, usage restrictions in the model weights) to prevent the model from being repurposed for malicious imitation or unauthorized surveillance, which is a standard expectation for powerful generative models.

Action items.

- **[science]** The ‘Ethical Considerations’ section (sec/6_limitations.tex) states that all human motion recordings are obtained with ‘informed consent’ but fails to specify the IRB approval number or the specific consent protocol used for the ‘large internally captured dataset’ mentioned in the Introduction. Given the use of human subjects for data collection, explicit IRB documentation is required to verify compliance with ethical standards.
- **[science]** The paper describes deploying a high-speed, zero-shot motion tracker on a physical

humanoid (Unitree-G1) capable of ‘collaboratively carrying boxes’ and executing ‘high-dynamic’ motions. The safety section mentions an ‘emergency stop mechanism’ but lacks a quantitative risk assessment or a detailed safety protocol for human-robot interaction (HRI) in unstructured environments, which is critical for preventing physical harm during real-world deployment.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling narrative on scaling data and model capacity for humanoid motion tracking, supported by extensive engineering and a large-scale dataset. However, from a strict scientific evidence perspective, the statistical robustness of the central claims requires strengthening.

First, the **scaling law analysis** (Sec. 5.1, lines 430-445) presents a clear trend in performance metrics (MPJPE, SR) as data increases from 2M to 2B tokens. However, the evidence is purely point-estimate based. There is no mention of the number of random seeds used for training, nor are there error bars, standard deviations, or confidence intervals reported in Table 2 or Fig. 4. In deep learning, performance can vary significantly between seeds, especially at the extremes of the scaling curve. Without reporting variance or conducting statistical significance tests (e.g., t-tests) between the 200M and 2B token configurations, the claim that the marginal gains are “real” rather than stochastic noise is weak. The authors must provide variance metrics or re-run experiments with multiple seeds to substantiate the scaling law.

Second, the **real-world generalization claim** relies on a very small test set. Table 3 (lines 280-310) reports quantitative metrics for only four specific dance sequences. While the qualitative videos (Fig. 3, Fig. 5) show a variety of motions, the quantitative evidence for “zero-shot generalization” is limited to these four clips. A sample size of four is statistically insufficient to claim robust generalization across “diverse, complex” unseen motions. The authors should either expand the real-world test set to include a larger, diverse collection of unseen motions (e.g., 20-50 clips across different categories) or explicitly frame the quantitative results as preliminary case studies rather than definitive proof of generalization.

Third, the **attribution of performance gains** to the Transformer architecture is confounded by the massive difference in training data. The baselines (GMT, TWIST, Any2Track) are trained on ~6-9M frames, whereas Humanoid-GPT is trained on 2B frames (Table 1, lines 130-150). The paper argues that the Transformer scales better than MLPs, but the primary comparison is between a 2B-frame Transformer and a 9M-frame MLP. To isolate the architectural contribution, the authors need an ablation study comparing a Transformer and an MLP trained on the *same* amount of data (e.g., both on 9M frames, or both on 2B frames if feasible). Without this, the claim that the Transformer is the primary driver of the performance leap, rather than the sheer

volume of data, remains an unproven hypothesis.

Finally, the **diversity metrics** (Sec. 4.2, lines 230-260) rely on HME embeddings and covariance volume. While the methodology is sound, the paper does not provide a baseline for what constitutes a “good” log-volume or gstd, nor does it correlate these specific diversity metrics directly with the final tracking success rate in a regression analysis. The claim that “diversity and balance are both necessary” is qualitative; a quantitative correlation would strengthen the evidence.

In summary, while the engineering achievement is significant, the scientific evidence supporting the specific claims about scaling laws, architectural superiority, and generalization robustness needs more statistical depth and controlled ablations.

Action items.

- **[science]** The scaling law analysis in Sec. 5.1 lacks statistical rigor. The text claims ‘marginal gains decrease’ but provides no confidence intervals, standard errors, or p-values for the differences between the 200M and 2B token runs. Without error bars or multiple seeds, the observed trend could be noise. Please report variance across seeds or statistical significance tests.
- **[science]** The real-world evaluation in Sec. 4.3 relies on only four specific dance motions for quantitative metrics. This sample size is insufficient to support the broad claim of ‘unprecedented zero-shot generalization’ to ‘diverse, complex’ motions. Please expand the test set to include a statistically significant number of unseen motion categories or provide a power analysis justifying the current sample size.
- **[science]** The comparison with baselines in Table 2 is confounded by data scale. Baselines are trained on ~6-9M frames, while Humanoid-GPT uses 2B frames. The paper attributes gains to architecture but does not isolate the effect by comparing a Transformer trained on 9M frames against an MLP trained on 9M frames. The claim that ‘Transformers offer superior tracking precision’ is not fully supported without this controlled ablation.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis presented in the manuscript is generally aligned with the narrative of scaling laws, but several critical gaps in reporting prevent full verification of the quantitative claims.

First, there is a discrepancy between the text and the provided tables regarding baseline performance. In Section 4.3, the authors state: “while larger models eventually reach competitive success rates (e.g., TCN-L achieves 89.05% at 2B tokens).” However, Table 1 (`tab:sim_scaling_backbone`) only reports results for MLP and TCN at 2M tokens, and for

Humanoid-GPT variants up to 2B tokens. The specific data point for TCN-L at 2B tokens is absent from the table. Without this data, the claim of saturation for TCN at high scales cannot be independently verified. The authors must either include this row in the table or remove the specific numerical claim from the text.

Second, the analysis of dataset diversity in Section 4.2 relies on point estimates for **gstd** and **log-volume** derived from a sample of 10,000 embeddings. The text claims a “4-5x increase” in log-volume for the curated dataset compared to AMASS. However, no measure of uncertainty (standard error, confidence intervals, or p-values from a statistical test) is provided. Given the high dimensionality of the embedding space and the potential for sampling bias, a statistical test (e.g., a permutation test or bootstrap confidence interval) is necessary to confirm that the observed difference is statistically significant and not an artifact of the sampling procedure.

Third, the “Scaling Laws” section (Section 5) makes strong claims about “predictable trends” and “monotonic scaling” but relies entirely on visual inspection of figures (Fig. 4, Fig. 5) which are not rendered in the text source. To meet the standard of a scaling law paper, the authors should explicitly report the fitted exponents (e.g., α in $Error \propto Data^{-\alpha}$) and their confidence intervals in the text or a supplementary table. Qualitative descriptions of curves are insufficient to establish a “law.”

Finally, while the ablation studies in the appendix vary hyperparameters (number of experts, history length), the results are presented as single-point comparisons without error bars or multiple random seeds. For claims regarding the “optimal” configuration (e.g., 384 experts), reporting the variance across seeds would strengthen the conclusion that the observed performance difference is robust.

Action items.

- **[science]** In Section 4.3 (Evaluation in Simulation), the text claims TCN-L achieves 89.05% SR at 2B tokens, but Table 1 (tab:sim_scaling_backbone) does not list a TCN-L row or a 2B token entry for TCN. This specific data point is missing from the provided evidence and must be added to the table or removed from the text to ensure reproducibility.
- **[science]** Section 4.2 defines diversity metrics (gstd, log-volume) but does not report the standard deviation or confidence intervals for these estimates across the 10,000 sampled embeddings. Given the claim that the new dataset has a ‘4-5x increase’ in log-volume, statistical significance testing or error bars are required to validate this difference is not due to sampling variance.
- **[science]** The scaling law analysis in Section 5 (Scaling Laws) describes trends qualitatively and references figures (Fig. 4, Fig. 5) that are not visible in the text source. The manuscript must explicitly state the fitted power-law exponents (e.g., $P \propto N^{-lpha}$) and their confidence intervals in the text or appendix to substantiate the claim of ‘predictable trends’.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript is generally well-written, with a clear narrative flow and strong technical exposition. The introduction effectively sets the stage for the “science of scale” argument, and the methodology is described with sufficient detail for a reader to understand the pipeline. However, there are several specific instances of grammatical errors, typos, and inconsistent phrasing that detract from the professional polish of the paper.

In Section 3.1, the phrase “presenting in Fig...” is grammatically awkward and should be revised to “as presented in Fig...” or “shown in Fig...”. Similarly, in Section 3.2, a run-on sentence involving a comma splice (“...temporal predicting, Moreover, because...”) disrupts the reading flow and requires splitting into two distinct sentences.

There are also minor inconsistencies in the text. In the “Metrics” paragraph of Section 4.2, the authors state they report “three quantitative metrics” but subsequently list five distinct metrics (SR, MPJPE, MPJVE, RootVelErr, MPKPE). This discrepancy should be corrected to “five” to avoid confusing the reader. Additionally, in the Appendix (Summary of Contributions), the term “video-esti data” appears to be a typo for “video-estimated data.”

Finally, while the use of the `\ding` command for listing items in the Introduction and Experiments sections is visually distinct, the phrasing in the “Metrics” section (e.g., “We report three quantitative metrics: : \textbf{Tracking Success Rate...}”) contains a double colon and slightly awkward punctuation that could be smoothed out. Addressing these specific line-level issues will significantly improve the overall readability and professionalism of the manuscript.

Action items.

- **[writing]** In Section 3.1 (Training Motion Experts), the text states ‘presenting in Fig. ef{fig:pipeline} extbf{(b)}’. This is grammatically incorrect and should be rephrased to ‘as presented in Fig. ef{fig:pipeline}(b)’ or ‘shown in Fig. ef{fig:pipeline}(b)’.
- **[writing]** In Section 3.2 (Building Zero-shot Foundational Tracker), the sentence ‘This design of our \methodname model naturally exploits the Transformer’s inherent strengths of parallel sequence supervision and autoregressive temporal predicting, Moreover, because...’ contains a comma splice and a double space. It should be split into two sentences or joined with a conjunction (e.g., ‘...predicting. Moreover, because...’).
- **[writing]** In Section 4.1 (Analysis on Data Diversity), the phrase ‘video-esti data’ appears in the Summary of Contributions (Appendix). This is a typo for ‘video-estimated data’ and should be corrected for clarity.
- **[writing]** In Section 4.2 (Evaluation in Simulation), the ‘Metrics’ paragraph lists five metrics but introduces them with ‘We report three quantitative metrics:’. This numerical inconsistency (three vs. five) is confusing and should be corrected to ‘five quantitative metrics’.